

Look at the special case when $A = B = C =$ square matrix. Then (A times A^2) is equal to (A^2 times A). The product in either order is A^3 . The matrix powers A^p follow the same rules as numbers:

$$A^p = \underbrace{A A A \cdots A}_{p \text{ factors}} \quad (A^p)(A^q) = A^{p+q} \quad (A^p)^q = A^{pq}.$$

Those are the ordinary laws for exponents. A^3 times A^4 is A^7 (seven factors). A^3 to the fourth power is A^{12} (twelve A 's). When p and q are zero or negative these rules still hold, provided A has a “ -1 power”—which is the *inverse matrix* A^{-1} . Then $A^0 = I$ is the identity matrix (no factors).

For a number, a^{-1} is $1/a$. For a matrix, the inverse is written A^{-1} . (It is *never* I/A , except this is allowed in MATLAB.) Every number has an inverse except $a = 0$. To decide when A has an inverse is a central problem in linear algebra. Section 2.5 will start on the answer. This section is a Bill of Rights for matrices, to say when A and B can be multiplied and how.

Block Matrices and Block Multiplication

We have to say one more thing about matrices. They can be cut into **blocks** (which are smaller matrices). This often happens naturally. Here is a 4 by 6 matrix broken into blocks of size 2 by 2—in this example each block is just I :

$$\begin{array}{l} \text{4 by 6 matrix} \\ \text{2 by 2 blocks} \end{array} \quad A = \left[\begin{array}{cc|cc|cc} 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{array} \right] = \begin{bmatrix} I & I & I \\ I & I & I \end{bmatrix}.$$

If B is also 4 by 6 and the block sizes match, you can add $A + B$ *a block at a time*.

We have seen block matrices before. The right side vector b was placed next to A in the “augmented matrix”. Then $[A \ b]$ has two blocks of different sizes. Multiplying by an elimination matrix gave $[EA \ Eb]$. No problem to multiply blocks times blocks, when their shapes permit.

Block multiplication If the cuts between columns of A match the cuts between rows of B , then block multiplication of AB is allowed:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \cdots \\ B_{21} & \cdots \end{bmatrix} = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & \cdots \\ A_{21}B_{11} + A_{22}B_{21} & \cdots \end{bmatrix}. \quad (1)$$

This equation is the same as if the blocks were numbers (which are 1 by 1 blocks). We are careful to keep A 's in front of B 's, because BA can be different.

Main point When matrices split into blocks, it is often simpler to see how they act. The block matrix of I 's above is much clearer than the original 4 by 6 matrix A .

Example 3 (Important special case) Let the blocks of A be its n columns. Let the blocks of B be its n rows. Then block multiplication AB adds up *columns times rows*:

$$\begin{array}{l} \text{Columns} \\ \text{times} \\ \text{rows} \end{array} \quad \left[\begin{array}{c|c|c} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{array} \right] \left[\begin{array}{c|c|c} - & b_1 & - \\ & \vdots & \\ - & b_n & - \end{array} \right] = \left[a_1 b_1 + \cdots + a_n b_n \right]. \quad (2)$$

This is another way to multiply matrices. Compare it with the usual rows times columns. Row 1 of A times column 1 of B gave the $(1, 1)$ entry in AB . Now *column 1 of A times row 1 of B* gives a full matrix—not just a single number. Look at this example:

$$\begin{array}{l} \text{Column 1 times row 1} \\ + \text{Column 2 times row 2} \end{array} \quad \left[\begin{array}{cc} 1 & 4 \\ 1 & 5 \end{array} \right] \left[\begin{array}{cc} 3 & 2 \\ 1 & 0 \end{array} \right] = \left[\begin{array}{c} 1 \\ 1 \end{array} \right] \left[\begin{array}{cc} 3 & 2 \end{array} \right] + \left[\begin{array}{c} 4 \\ 5 \end{array} \right] \left[\begin{array}{cc} 1 & 0 \end{array} \right] = \left[\begin{array}{cc} 3 & 2 \\ 3 & 2 \end{array} \right] + \left[\begin{array}{cc} 4 & 0 \\ 5 & 0 \end{array} \right]. \quad (3)$$

We stop there so you can see columns multiplying rows. If a 2 by 1 matrix (a column) multiplies a 1 by 2 matrix (a row), the result is 2 by 2. That is what we found. Dot products are *inner* products and these are *outer* products. In the top left corner the answer is $3 + 4 = 7$. This agrees with the row-column dot product of $(1, 4)$ with $(3, 1)$.

Summary The usual way, rows times columns, gives four dot products (8 multiplications). The new way, columns times rows, gives two full matrices (the same 8 multiplications). The 8 multiplications, and the 4 additions, are just executed in a different order.

Example 4 (Elimination by blocks) Suppose the first column of A contains 1, 3, 4. To change 3 and 4 to 0 and 0, multiply the pivot row by 3 and 4 and subtract. Those row operations are really multiplications by elimination matrices E_{21} and E_{31} :

$$\text{One at a time} \quad E_{21} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad \text{and} \quad E_{31} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{array} \right].$$

The “block idea” is to do both eliminations with one matrix E . That matrix clears out the whole first column of A below the pivot $a = 1$:

$$E = \left[\begin{array}{ccc} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -4 & 0 & 1 \end{array} \right] \quad \text{multiplies} \quad \left[\begin{array}{ccc} 1 & x & x \\ 3 & x & x \\ 4 & x & x \end{array} \right] \quad \text{to give} \quad EA = \left[\begin{array}{ccc} 1 & x & x \\ 0 & x & x \\ 0 & x & x \end{array} \right].$$

Using inverses from 2.5, a block matrix E can do elimination on a whole (block) column of A . Suppose A has four blocks A, B, C, D . Watch how E multiplies A by blocks:

$$\begin{array}{l} \text{Block} \\ \text{elimination} \end{array} \quad \left[\begin{array}{c|c} I & 0 \\ \hline -CA^{-1} & I \end{array} \right] \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \left[\begin{array}{c|c} A & B \\ \hline 0 & D - CA^{-1}B \end{array} \right]. \quad (4)$$

Elimination multiplies the first row $[A \ B]$ by CA^{-1} (previously c/a). It subtracts from C to get a zero block in the first column. It subtracts from D to get $S = D - CA^{-1}B$.

This is ordinary elimination, a column at a time—written in blocks. That final block S is $D - CA^{-1}B$, just like $d - cb/a$. This is called the *Schur complement*.

■ REVIEW OF THE KEY IDEAS ■

1. The (i, j) entry of AB is (row i of A) $\cdot$ (column j of B).
2. An m by n matrix times an n by p matrix uses mnp separate multiplications.
3. A times BC equals AB times C (surprisingly important).
4. AB is also the sum of these matrices: (column j of A) times (row j of B).
5. Block multiplication is allowed when the block shapes match correctly.
6. Block elimination produces the *Schur complement* $D - CA^{-1}B$.

■ WORKED EXAMPLES ■

2.4 A Put yourself in the position of the author! I want to show you matrix multiplications that are *special*, but mostly I am stuck with small matrices. There is one terrific family of **Pascal matrices**, and they come in all sizes, and above all they have real meaning. I think 4 by 4 is a good size to show some of their amazing patterns.

Here is the lower triangular Pascal matrix L . Its entries come from “*Pascal’s triangle*”. I will multiply L times the **ones** vector, and the **powers** vector:

$$\text{Pascal matrix } \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \\ 8 \end{bmatrix} \quad \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1+x \\ (1+x)^2 \\ (1+x)^3 \end{bmatrix}.$$

Each row of L leads to the next row: *Add an entry to the one on its left to get the entry below.* In symbols $\ell_{ij} + \ell_{i,j-1} = \ell_{i+1,j}$. The numbers after 1, 3, 3, 1 would be 1, 4, 6, 4, 1. Pascal lived in the 1600’s, long before matrices, but his triangle fits perfectly into L .

Multiplying by **ones** is the same as adding up each row, to get powers of 2. By writing out L times **powers of x** , you see the entries of L as the “binomial coefficients” that are so essential to gamblers:

$$1 + 2x + 1x^2 = (1+x)^2 \quad 1 + 3x + 3x^2 + 1x^3 = (1+x)^3$$

The number “3” counts the ways to get Heads once and Tails twice in three coin flips: HTT and THT and TTH. The other “3” counts the ways to get Heads twice: HHT and

HTH and THH. Those are examples of “ i choose j ” = the number of ways to get j heads in i coin flips. That number is exactly ℓ_{ij} , if we start counting rows and columns of L at $i = 0$ and $j = 0$ (and remember $0! = 1$):

$$\ell_{ij} = \binom{i}{j} = i \text{ choose } j = \frac{i!}{j!(i-j)!} \quad \binom{4}{2} = \frac{4!}{2!2!} = \frac{24}{(2)(2)} = 6$$

There are six ways to choose two aces out of four aces. We will see Pascal’s triangle and these matrices again. Here are the questions I want to ask now:

1. What is $H = L^2$? This is the “hypercube matrix”.
2. Multiply H times ones and powers.
3. The last row of H is 8, 12, 6, 1. A cube has 8 corners, 12 edges, 6 faces, 1 box. What would the next row of H tell about a hypercube in 4D?

Solution Multiply L times L to get the hypercube matrix $H = L^2$:

$$\begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 2 & 1 & \\ 1 & 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 4 & 4 & 1 & \\ 8 & 12 & 6 & 1 \end{bmatrix} = H.$$

Now multiply H times the vectors of ones and powers:

$$\begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 4 & 4 & 1 & \\ 8 & 12 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \\ 27 \end{bmatrix} \quad \begin{bmatrix} 1 & & & \\ 2 & 1 & & \\ 4 & 4 & 1 & \\ 8 & 12 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2+x \\ (2+x)^2 \\ (2+x)^3 \end{bmatrix}$$

If $x = 1$ we get the powers of 3. If $x = 0$ we get powers of 2. When L produces powers of $1 + x$, applying L again produces powers of $2 + x$.

How do the rows of H count corners and edges and faces of a cube? A square in 2D has 4 corners, 4 edges, 1 face. Add one dimension at a time:

Connect two squares to get a 3D cube. Connect two cubes to get a 4D hypercube.

The cube has 8 corners and 12 edges: 4 edges in each square and 4 between the squares. The cube has 6 faces: 1 in each square and 4 faces between the squares. This row 8, 12, 6, 1 will lead to the next row 16, 32, 24, 8, 1. The rule is $2h_{ij} + h_{i,j-1} = h_{i+1,j}$.

Can you see this in four dimensions? The hypercube has 16 corners, no problem. It has 12 edges from one cube, 12 from the other cube, 8 that connect corners of those cubes: total 32 edges. It has 6 faces from each separate cube and 12 more from connecting pairs of edges: total $2 \times 6 + 12 = 24$ faces. It has one box from each cube and 6 more from connecting pairs of faces: total 8 boxes. And finally 1 hypercube.

2.4 B For these matrices, when does $AB = BA$? When does $BC = CB$? When does A times BC equal AB times C ? Give the conditions on their entries p, q, r, z :

$$A = \begin{bmatrix} p & 0 \\ q & r \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & z \\ 0 & 0 \end{bmatrix}$$

If $p, q, r, 1, z$ are 4 by 4 blocks instead of numbers, do the answers change?

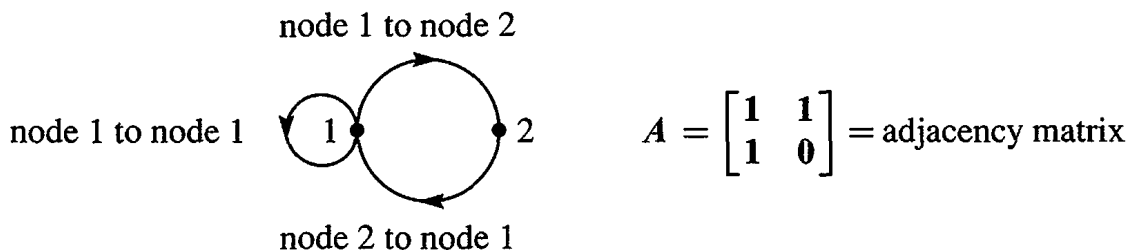
Solution First of all, A times BC *always* equals AB times C . Parentheses are not needed in $A(BC) = (AB)C = ABC$. But we must keep the matrices in this order:

$$\text{Usually } AB \neq BA \quad AB = \begin{bmatrix} p & p \\ q & q+r \end{bmatrix} \quad BA = \begin{bmatrix} p+q & r \\ q & r \end{bmatrix}.$$

$$\text{By chance } BC = CB \quad BC = \begin{bmatrix} 0 & z \\ 0 & 0 \end{bmatrix} \quad CB = \begin{bmatrix} 0 & z \\ 0 & 0 \end{bmatrix}.$$

B and C happen to commute. Part of the explanation is that the diagonal of B is I , which commutes with all 2 by 2 matrices. When p, q, r, z are 4 by 4 blocks and 1 changes to I , all these products remain correct. So the answers are the same.

2.4 C A **directed graph** starts with n nodes. The n by n **adjacency matrix** has $a_{ij} = 1$ when an edge leaves node i and enters node j ; if no edge then $a_{ij} = 0$.



The i, j entry of A^2 is $\sum a_{ik}a_{kj}$. This is $a_{i1}a_{1j} + \cdots + a_{in}a_{nj}$. Why does that sum count the *two-step paths* from i to any node to j ? The i, j entry of A^k counts k -step paths:

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{Count paths with two edges} \quad \begin{bmatrix} 1 \text{ to } 2 \text{ to } 1, 1 \text{ to } 1 \text{ to } 1 & 1 \text{ to } 1 \text{ to } 2 \\ 2 \text{ to } 1 \text{ to } 1 & 2 \text{ to } 1 \text{ to } 2 \end{bmatrix}$$

List all of the 3-step paths between each pair of nodes and compare with A^3 .

Solution The number $a_{ik}a_{kj}$ will be “1” if there is an edge from node i to k and an edge from k to j . This is a 2-step path. The number $a_{ik}a_{kj}$ will be “0” if either of those edges (i to k , k to j) is missing. So the sum of $a_{ik}a_{kj}$ is the number of 2-step paths leaving i and entering j . Matrix multiplication is just right for this count.

The 3-step paths are counted by A^3 ; we look at paths to node 2:

$$A^3 = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \quad \begin{array}{l} \text{counts the paths} \\ \text{with three steps} \end{array} \quad \begin{bmatrix} \cdots & 1 \text{ to } 1 \text{ to } 1 \text{ to } 2, 1 \text{ to } 2 \text{ to } 1 \text{ to } 2 \\ \cdots & 2 \text{ to } 1 \text{ to } 1 \text{ to } 2 \end{bmatrix}$$

These A^k contain the Fibonacci numbers 0, 1, 1, 2, 3, 5, 8, 13, ... coming in Section 6.2. Multiplying A by A^k involves Fibonacci's rule $F_{k+2} = F_{k+1} + F_k$ (as in $13 = 8 + 5$):

$$(A)(A^k) = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{bmatrix} = \begin{bmatrix} F_{k+2} & F_{k+1} \\ F_{k+1} & F_k \end{bmatrix} = A^{k+1}.$$

There are 13 six-step paths from node 1 to node 1, but I can't find them all.

A^k also counts words. A path like 1 to 1 to 2 to 1 corresponds to the word **aaba**. The letter **b** can't repeat because there is no edge from 2 to 2. The i, j entry of A^k counts the words of length $k + 1$ that start with the i th letter and end with the j th.

Problem Set 2.4

Problems 1–16 are about the laws of matrix multiplication.

- 1** A is 3 by 5, B is 5 by 3, C is 5 by 1, and D is 3 by 1. All entries are 1. Which of these matrix operations are allowed, and what are the results?

$$BA \qquad AB \qquad ABD \qquad DBA \qquad A(B + C).$$

- 2** What rows or columns or matrices do you multiply to find

- (a) the third column of AB ?
- (b) the first row of AB ?
- (c) the entry in row 3, column 4 of AB ?
- (d) the entry in row 1, column 1 of CDE ?

- 3** Add AB to AC and compare with $A(B + C)$:

$$A = \begin{bmatrix} 1 & 5 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 2 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}.$$

- 4** In Problem 3, multiply A times BC . Then multiply AB times C .

- 5 Compute A^2 and A^3 . Make a prediction for A^5 and A^n :

$$A = \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix}.$$

- 6 Show that $(A + B)^2$ is different from $A^2 + 2AB + B^2$, when

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}.$$

Write down the correct rule for $(A + B)(A + B) = A^2 + \underline{\hspace{2cm}} + B^2$.

- 7 True or false. Give a specific example when false:

- (a) If columns 1 and 3 of B are the same, so are columns 1 and 3 of AB .
- (b) If rows 1 and 3 of B are the same, so are rows 1 and 3 of AB .
- (c) If rows 1 and 3 of A are the same, so are rows 1 and 3 of ABC .
- (d) $(AB)^2 = A^2B^2$.

- 8 How is each row of DA and EA related to the rows of A , when

$$D = \begin{bmatrix} 3 & 0 \\ 0 & 5 \end{bmatrix} \quad \text{and} \quad E = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$

How is each column of AD and AE related to the columns of A ?

- 9 Row 1 of A is added to row 2. This gives EA below. Then column 1 of EA is added to column 2 to produce $(EA)F$:

$$EA = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix}$$

$$\text{and} \quad (EA)F = (EA) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & a+b \\ a+c & a+c+b+d \end{bmatrix}.$$

- (a) Do those steps in the opposite order. First add column 1 of A to column 2 by AF , then add row 1 of AF to row 2 by $E(AF)$.
 - (b) Compare with $(EA)F$. What law is obeyed by matrix multiplication?
- 10 Row 1 of A is again added to row 2 to produce EA . Then F adds row 2 of EA to row 1. The result is $F(EA)$:

$$F(EA) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix} = \begin{bmatrix} 2a+c & 2b+d \\ a+c & b+d \end{bmatrix}.$$

- (a) Do those steps in the opposite order: first add row 2 to row 1 by FA , then add row 1 of FA to row 2.
- (b) What law is or is not obeyed by matrix multiplication?

11 (3 by 3 matrices) Choose the only B so that for every matrix A

- (a) $BA = 4A$
- (b) $BA = 4B$
- (c) BA has rows 1 and 3 of A reversed and row 2 unchanged
- (d) All rows of BA are the same as row 1 of A .

12 Suppose $AB = BA$ and $AC = CA$ for these two particular matrices B and C :

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ commutes with } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Prove that $a = d$ and $b = c = 0$. Then A is a multiple of I . The only matrices that commute with B and C and all other 2 by 2 matrices are $A = \text{multiple of } I$.

13 Which of the following matrices are guaranteed to equal $(A - B)^2$: $A^2 - B^2$, $(B - A)^2$, $A^2 - 2AB + B^2$, $A(A - B) - B(A - B)$, $A^2 - AB - BA + B^2$?

14 True or false:

- (a) If A^2 is defined then A is necessarily square.
- (b) If AB and BA are defined then A and B are square.
- (c) If AB and BA are defined then AB and BA are square.
- (d) If $AB = B$ then $A = I$.

15 If A is m by n , how many separate multiplications are involved when

- (a) A multiplies a vector x with n components?
- (b) A multiplies an n by p matrix B ?
- (c) A multiplies itself to produce A^2 ? Here $m = n$.

16 For $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 4 \\ 1 & 0 & 6 \end{bmatrix}$, compute these answers *and nothing more*:

- (a) column 2 of AB
- (b) row 2 of AB
- (c) row 2 of $AA = A^2$
- (d) row 2 of $AAA = A^3$.

Problems 17–19 use a_{ij} for the entry in row i , column j of A .

17 Write down the 3 by 3 matrix A whose entries are

- (a) $a_{ij} = \text{minimum of } i \text{ and } j$
- (b) $a_{ij} = (-1)^{i+j}$
- (c) $a_{ij} = i/j$.

- 18** What words would you use to describe each of these classes of matrices? Give a 3 by 3 example in each class. Which matrix belongs to all four classes?

(a) $a_{ij} = 0$ if $i \neq j$

(b) $a_{ij} = 0$ if $i < j$

(c) $a_{ij} = a_{ji}$

(d) $a_{ij} = a_{1j}$.

- 19** The entries of A are a_{ij} . Assuming that zeros don't appear, what is

(a) the first pivot?

(b) the multiplier ℓ_{31} of row 1 to be subtracted from row 3?

(c) the new entry that replaces a_{32} after that subtraction?

(d) the second pivot?

Problems 20–24 involve powers of A .

- 20** Compute A^2 , A^3 , A^4 and also Av , A^2v , A^3v , A^4v for

$$A = \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad v = \begin{bmatrix} x \\ y \\ z \\ t \end{bmatrix}.$$

- 21** Find all the powers A^2 , A^3 , $\dots$ and AB , $(AB)^2$, $\dots$ for

$$A = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

- 22** By trial and error find real nonzero 2 by 2 matrices such that

$$A^2 = -I \quad BC = 0 \quad DE = -ED \text{ (not allowing } DE = 0\text{)}.$$

- 23** (a) Find a nonzero matrix A for which $A^2 = 0$.
 (b) Find a matrix that has $A^2 \neq 0$ but $A^3 = 0$.

- 24** By experiment with $n = 2$ and $n = 3$ predict A^n for these matrices:

$$A_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad A_3 = \begin{bmatrix} a & b \\ 0 & 0 \end{bmatrix}.$$

Problems 25–31 use column-row multiplication and block multiplication.

25 Multiply A times I using columns of A (3 by 3) times rows of I .

26 Multiply AB using columns times rows:

$$AB = \begin{bmatrix} 1 & 0 \\ 2 & 4 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 0 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} [3 \ 3 \ 0] + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}.$$

27 Show that the product of upper triangular matrices is always upper triangular:

$$AB = \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} \begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \end{bmatrix} = \begin{bmatrix} & & \\ 0 & & \\ 0 & 0 & \end{bmatrix}.$$

Proof using dot products (Row times column) (Row 2 of A) $\cdot$ (column 1 of B) = 0. Which other dot products give zeros?

Proof using full matrices (Column times row) Draw x 's and 0's in (column 2 of A) times (row 2 of B). Also show (column 3 of A) times (row 3 of B).

28 Draw the cuts in A (2 by 3) and B (3 by 4) and AB to show how each of the four multiplication rules is really a block multiplication:

- | | |
|--|---|
| (1) Matrix A times columns of B . | Columns of AB |
| (2) Rows of A times the matrix B . | Rows of AB |
| (3) Rows of A times columns of B . | Inner products (numbers in AB) |
| (4) Columns of A times rows of B . | Outer products (matrices add to AB) |

29 Which matrices E_{21} and E_{31} produce zeros in the (2, 1) and (3, 1) positions of $E_{21}A$ and $E_{31}A$?

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -2 & 0 & 1 \\ 8 & 5 & 3 \end{bmatrix}.$$

Find the single matrix $E = E_{31}E_{21}$ that produces both zeros at once. Multiply EA .

30 Block multiplication says that column 1 is eliminated by

$$EA = \begin{bmatrix} 1 & 0 \\ -c/a & I \end{bmatrix} \begin{bmatrix} a & b \\ c & D \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & D - cb/a \end{bmatrix}.$$

In Problem 29, what are c and D and what is $D - cb/a$?

31 With $i^2 = -1$, the product of $(A + iB)$ and $(x + iy)$ is $Ax + iBx + iAy - By$. Use blocks to separate the real part without i from the imaginary part that multiplies i :

$$\begin{bmatrix} A & -B \\ ? & ? \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} Ax - By \\ ? \end{bmatrix} \begin{matrix} \text{real part} \\ \text{imaginary part} \end{matrix}$$

- 32 (Very important) Suppose you solve $Ax = b$ for three special right sides b :

$$Ax_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad Ax_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad Ax_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

If the three solutions x_1, x_2, x_3 are the columns of a matrix X , what is A times X ?

- 33 If the three solutions in Question 32 are $x_1 = (1, 1, 1)$ and $x_2 = (0, 1, 1)$ and $x_3 = (0, 0, 1)$, solve $Ax = b$ when $b = (3, 5, 8)$. Challenge problem: What is A ?
- 34 Find all matrices $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ that satisfy $A \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} A$.
- 35 Suppose a “circle graph” has 4 nodes connected (in both directions) by edges around a circle. What is its adjacency matrix from Worked Example 2.4 C? What is A^2 ? Find all the 2-step paths (or 3-letter words) predicted by A^2 .

Challenge Problems

- 36 **Practical question** Suppose A is m by n , B is n by p , and C is p by q . Then the multiplication count for $(AB)C$ is $mnp + mpq$. The same answer comes from A times BC with $mnq + npq$ separate multiplications. Notice npq for BC .
- (a) If A is 2 by 4, B is 4 by 7, and C is 7 by 10, do you prefer $(AB)C$ or $A(BC)$?
- (b) With N -component vectors, would you choose $(u^T v)w^T$ or $u^T(vw^T)$?
- (c) Divide by $mnpq$ to show that $(AB)C$ is faster when $n^{-1} + q^{-1} < m^{-1} + p^{-1}$.
- 37 To prove that $(AB)C = A(BC)$, use the column vectors $b_1, \dots, b_n$ of B . First suppose that C has only one column c with entries $c_1, \dots, c_n$:
- AB has columns $Ab_1, \dots, Ab_n$ and then $(AB)c$ equals $c_1 Ab_1 + \dots + c_n Ab_n$.
- Bc has one column $c_1 b_1 + \dots + c_n b_n$ and then $A(Bc)$ equals $A(c_1 b_1 + \dots + c_n b_n)$.
- Linearity gives equality of those two sums. This proves $(AB)c = A(Bc)$. The same is true for all other _____ of C . Therefore $(AB)C = A(BC)$. Apply to inverses: If $BA = I$ and $AC = I$, prove that the left-inverse B equals the right-inverse C .

2.5 Inverse Matrices

Suppose A is a square matrix. We look for an “*inverse matrix*” A^{-1} of the same size, such that A^{-1} times A equals I . Whatever A does, A^{-1} undoes. Their product is the identity matrix—which does nothing to a vector, so $A^{-1}Ax = x$. But A^{-1} might not exist.

What a matrix mostly does is to multiply a vector x . Multiplying $Ax = b$ by A^{-1} gives $A^{-1}Ax = A^{-1}b$. **This is $x = A^{-1}b$.** The product $A^{-1}A$ is like multiplying by a number and then dividing by that number. A number has an inverse if it is not zero—matrices are more complicated and more interesting. The matrix A^{-1} is called “ A inverse.”

DEFINITION The matrix A is *invertible* if there exists a matrix A^{-1} such that

$$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I. \quad (1)$$

Not all matrices have inverses. This is the first question we ask about a square matrix: Is A invertible? We don’t mean that we immediately calculate A^{-1} . In most problems we never compute it! Here are six “notes” about A^{-1} .

Note 1 *The inverse exists if and only if elimination produces n pivots* (row exchanges are allowed). Elimination solves $Ax = b$ without explicitly using the matrix A^{-1} .

Note 2 The matrix A cannot have two different inverses. Suppose $BA = I$ and also $AC = I$. Then $B = C$, according to this “proof by parentheses”:

$$B(AC) = (BA)C \quad \text{gives} \quad BI = IC \quad \text{or} \quad B = C. \quad (2)$$

This shows that a *left-inverse* B (multiplying from the left) and a *right-inverse* C (multiplying A from the right to give $AC = I$) must be the *same matrix*.

Note 3 If A is invertible, the one and only solution to $Ax = b$ is $x = A^{-1}b$:

$$\text{Multiply } Ax = b \text{ by } A^{-1}. \text{ Then } x = A^{-1}Ax = A^{-1}b.$$

Note 4 (Important) *Suppose there is a nonzero vector x such that $Ax = 0$. Then A cannot have an inverse.* No matrix can bring 0 back to x .

If A is invertible, then $Ax = 0$ can only have the zero solution $x = A^{-1}0 = 0$.

Note 5 A 2 by 2 matrix is invertible if and only if $ad - bc$ is not zero:

$$\text{2 by 2 Inverse:} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \quad (3)$$

This number $ad - bc$ is the *determinant* of A . A matrix is invertible if its determinant is not zero (Chapter 5). The test for n pivots is usually decided before the determinant appears.

Note 6 A diagonal matrix has an inverse provided no diagonal entries are zero:

$$\text{If } A = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} 1/d_1 & & \\ & \ddots & \\ & & 1/d_n \end{bmatrix}.$$

Example 1 The 2 by 2 matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$ is not invertible. It fails the test in Note 5, because $ad - bc$ equals $2 - 2 = 0$. It fails the test in Note 3, because $Ax = \mathbf{0}$ when $x = (2, -1)$. It fails to have two pivots as required by Note 1.

Elimination turns the second row of this matrix A into a zero row.

The Inverse of a Product AB

For two nonzero numbers a and b , the sum $a + b$ might or might not be invertible. The numbers $a = 3$ and $b = -3$ have inverses $\frac{1}{3}$ and $-\frac{1}{3}$. Their sum $a + b = 0$ has no inverse. But the product $ab = -9$ does have an inverse, which is $\frac{1}{3}$ times $-\frac{1}{3}$.

For two matrices A and B , the situation is similar. It is hard to say much about the invertibility of $A + B$. But the *product* AB has an inverse, if and only if the two factors A and B are separately invertible (and the same size). The important point is that A^{-1} and B^{-1} come in *reverse order*:

If A and B are invertible then so is AB . The inverse of a product AB is

$$(AB)^{-1} = B^{-1}A^{-1}. \quad (4)$$

To see why the order is reversed, multiply AB times $B^{-1}A^{-1}$. Inside that is $BB^{-1} = I$:

$$\text{Inverse of } AB \quad (AB)(B^{-1}A^{-1}) = AIA^{-1} = AA^{-1} = I.$$

We moved parentheses to multiply BB^{-1} first. Similarly $B^{-1}A^{-1}$ times AB equals I . This illustrates a basic rule of mathematics: Inverses come in reverse order. It is also common sense: If you put on socks and then shoes, the first to be taken off are the _____. The same reverse order applies to three or more matrices:

$$\text{Reverse order} \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1}. \quad (5)$$

Example 2 *Inverse of an elimination matrix.* If E subtracts 5 times row 1 from row 2, then E^{-1} adds 5 times row 1 to row 2:

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiply EE^{-1} to get the identity matrix I . Also multiply $E^{-1}E$ to get I . We are adding and subtracting the same 5 times row 1. Whether we add and then subtract (this is EE^{-1}) or subtract and then add (this is $E^{-1}E$), we are back at the start.

For square matrices, an inverse on one side is automatically an inverse on the other side. If $AB = I$ then automatically $BA = I$. In that case B is A^{-1} . This is very useful to know but we are not ready to prove it.

Example 3 Suppose F subtracts 4 times row 2 from row 3, and F^{-1} adds it back:

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \quad \text{and} \quad F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

Now multiply F by the matrix E in Example 2 to find FE . Also multiply E^{-1} times F^{-1} to find $(FE)^{-1}$. Notice the orders FE and $E^{-1}F^{-1}$!

$$FE = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 20 & -4 & 1 \end{bmatrix} \quad \text{is inverted by} \quad E^{-1}F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}. \quad (6)$$

The result is beautiful and correct. The product FE contains “20” but its inverse doesn’t. E subtracts 5 times row 1 from row 2. Then F subtracts 4 times the *new* row 2 (changed by row 1) from row 3. **In this order FE , row 3 feels an effect from row 1.**

In the order $E^{-1}F^{-1}$, that effect does not happen. First F^{-1} adds 4 times row 2 to row 3. After that, E^{-1} adds 5 times row 1 to row 2. There is no 20, because row 3 doesn’t change again. **In this order $E^{-1}F^{-1}$, row 3 feels no effect from row 1.**

In elimination order F follows E . In reverse order E^{-1} follows F^{-1} .

$E^{-1}F^{-1}$ is quick. The multipliers 5, 4 fall into place below the diagonal of 1’s.

This special multiplication $E^{-1}F^{-1}$ and $E^{-1}F^{-1}G^{-1}$ will be useful in the next section. We will explain it again, more completely. In this section our job is A^{-1} , and we expect some serious work to compute it. Here is a way to organize that computation.

Calculating A^{-1} by Gauss-Jordan Elimination

I hinted that A^{-1} might not be explicitly needed. The equation $Ax = b$ is solved by $x = A^{-1}b$. But it is not necessary or efficient to compute A^{-1} and multiply it times b . *Elimination goes directly to x .* Elimination is also the way to calculate A^{-1} , as we now show. The Gauss-Jordan idea is to solve $AA^{-1} = I$, *finding each column of A^{-1} .*

A multiplies the first column of A^{-1} (call that x_1) to give the first column of I (call that e_1). This is our equation $Ax_1 = e_1 = (1, 0, 0)$. There will be two more equations. Each of the columns x_1, x_2, x_3 of A^{-1} is multiplied by A to produce a column of I :

$$\text{3 columns of } A^{-1} \quad AA^{-1} = A[x_1 \ x_2 \ x_3] = [e_1 \ e_2 \ e_3] = I. \quad (7)$$

To invert a 3 by 3 matrix A , we have to solve three systems of equations: $Ax_1 = e_1$ and $Ax_2 = e_2 = (0, 1, 0)$ and $Ax_3 = e_3 = (0, 0, 1)$. Gauss-Jordan finds A^{-1} this way.

The **Gauss-Jordan method** computes A^{-1} by solving *all n equations together*. Usually the “augmented matrix” $[A \ b]$ has one extra column b . Now we have three right sides e_1, e_2, e_3 (when A is 3 by 3). They are the columns of I , so the augmented matrix is really the block matrix $[A \ I]$. I take this chance to invert my favorite matrix K , with 2's on the main diagonal and -1 's next to the 2's:

$$\begin{aligned}
 [K \ e_1 \ e_2 \ e_3] &= \begin{bmatrix} 2 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{bmatrix} && \text{Start Gauss-Jordan on } K \\
 &\rightarrow \begin{bmatrix} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{bmatrix} && (\frac{1}{2} \text{ row } 1 + \text{row } 2) \\
 &\rightarrow \begin{bmatrix} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -1 & \frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{bmatrix} && (\frac{2}{3} \text{ row } 2 + \text{row } 3)
 \end{aligned}$$

We are halfway to K^{-1} . The matrix in the first three columns is U (upper triangular). The pivots $2, \frac{3}{2}, \frac{4}{3}$ are on its diagonal. Gauss would finish by back substitution. The contribution of Jordan is *to continue with elimination!* He goes all the way to the “**reduced echelon form**”. Rows are added to rows above them, to produce *zeros above the pivots*:

$$\begin{aligned}
 \left(\begin{array}{l} \text{Zero above} \\ \text{third pivot} \end{array} \right) &\rightarrow \begin{bmatrix} 2 & -1 & 0 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{bmatrix} && (\frac{3}{4} \text{ row } 3 + \text{row } 2) \\
 \left(\begin{array}{l} \text{Zero above} \\ \text{second pivot} \end{array} \right) &\rightarrow \begin{bmatrix} 2 & 0 & 0 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & \frac{3}{2} & 0 & \frac{3}{4} & \frac{3}{2} & \frac{3}{4} \\ 0 & 0 & \frac{4}{3} & \frac{1}{3} & \frac{2}{3} & 1 \end{bmatrix} && (\frac{2}{3} \text{ row } 2 + \text{row } 1)
 \end{aligned}$$

The last Gauss-Jordan step is to divide each row by its pivot. The new pivots are 1. We have reached I in the first half of the matrix, because K is invertible. *The three columns of K^{-1} are in the second half of $[I \ K^{-1}]$:*

$$\begin{aligned}
 & \begin{array}{l} \text{(divide by 2)} \\ \text{(divide by } \frac{3}{2}) \\ \text{(divide by } \frac{4}{3}) \end{array} \begin{bmatrix} 1 & 0 & 0 & \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 & \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{bmatrix} = [I \ x_1 \ x_2 \ x_3] = [I \ K^{-1}].
 \end{aligned}$$

Starting from the 3 by 6 matrix $[K \ I]$, we ended with $[I \ K^{-1}]$. Here is the whole Gauss-Jordan process on one line for any invertible matrix A :

Gauss-Jordan

Multiply $[A \ I]$ by A^{-1} to get $[I \ A^{-1}]$.

The elimination steps create the inverse matrix while changing A to I . For large matrices, we probably don't want A^{-1} at all. But for small matrices, it can be very worthwhile to know the inverse. We add three observations about this particular K^{-1} because it is an important example. We introduce the words *symmetric*, *tridiagonal*, and *determinant*:

1. K is *symmetric* across its main diagonal. So is K^{-1} .
2. K is *tridiagonal* (only three nonzero diagonals). But K^{-1} is a dense matrix with no zeros. That is another reason we don't often compute inverse matrices. The inverse of a band matrix is generally a dense matrix.
3. The *product of pivots* is $2(\frac{3}{2})(\frac{4}{3}) = 4$. This number 4 is the *determinant* of K .

$$K^{-1} \text{ involves division by the determinant} \quad K^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}. \quad (8)$$

This is why an invertible matrix cannot have a zero determinant.

Example 4 Find A^{-1} by Gauss-Jordan elimination starting from $A = \begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$. There are two row operations and then a division to put 1's in the pivots:

$$\begin{aligned} [A \ I] &= \begin{bmatrix} 2 & 3 & 1 & 0 \\ 4 & 7 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} \quad (\text{this is } [U \ L^{-1}]) \\ &\rightarrow \begin{bmatrix} 2 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \frac{7}{2} & -\frac{3}{2} \\ 0 & 1 & -2 & 1 \end{bmatrix} \quad (\text{this is } [I \ A^{-1}]). \end{aligned}$$

That A^{-1} involves division by the determinant $ad - bc = 2 \cdot 7 - 3 \cdot 4 = 2$. The code for $X = \text{inverse}(A)$ can use **rref**, the "row reduced echelon form" from Chapter 3:

```
I = eye(n);           % Define the n by n identity matrix
R = rref([A I]);       % Eliminate on the augmented matrix [A I]
X = R(:, n+1:n+1+n);  % Pick A^{-1} from the last n columns of R
```

A must be invertible, or elimination cannot reduce it to I (in the left half of R).

Gauss-Jordan shows why A^{-1} is expensive. We must solve n equations for its n columns.

To solve $Ax = b$ without A^{-1} , we deal with *one* column b to find one column x .

In defense of A^{-1} , we want to say that its cost is not n times the cost of one system $Ax = b$. Surprisingly, the cost for n columns is only multiplied by 3. This saving is because the n equations $Ax_i = e_i$ all involve the same matrix A . Working with the right sides is relatively cheap, because elimination only has to be done once on A .

The complete A^{-1} needs n^3 elimination steps, where a single x needs $n^3/3$. The next section calculates these costs.

Singular versus Invertible

We come back to the central question. Which matrices have inverses? The start of this section proposed the pivot test: *A^{-1} exists exactly when A has a full set of n pivots.* (Row exchanges are allowed.) Now we can prove that by Gauss-Jordan elimination:

1. With n pivots, elimination solves all the equations $Ax_i = e_i$. The columns x_i go into A^{-1} . Then $AA^{-1} = I$ and A^{-1} is at least a *right-inverse*.
2. Elimination is really a sequence of multiplications by E 's and P 's and D^{-1} :

$$\text{Left-inverse} \quad (D^{-1} \dots E \dots P \dots E)A = I. \quad (9)$$

D^{-1} divides by the pivots. The matrices E produce zeros below and above the pivots. P will exchange rows if needed (see Section 2.7). The product matrix in equation (9) is evidently a *left-inverse*. With n pivots we have reached $A^{-1}A = I$.

The right-inverse equals the left-inverse. That was Note 2 at the start of in this section. So a square matrix with a full set of pivots will always have a two-sided inverse.

Reasoning in reverse will now show that *A must have n pivots if $AC = I$.* (Then we deduce that C is also a left-inverse and $CA = I$.) Here is one route to those conclusions:

1. If A doesn't have n pivots, elimination will lead to a *zero row*.
2. Those elimination steps are taken by an invertible M . *So a row of MA is zero.*
3. If $AC = I$ had been possible, then $MAC = M$. The zero row of MA , times C , gives a zero row of M itself.
4. An invertible matrix M can't have a zero row! *A must have n pivots if $AC = I$.*

That argument took four steps, but the outcome is short and important.

Elimination gives a complete test for invertibility of a square matrix. *A^{-1} exists (and Gauss-Jordan finds it) exactly when A has n pivots.* The argument above shows more:

$$\text{If } AC = I \text{ then } CA = I \text{ and } C = A^{-1}$$

Example 5 If L is lower triangular with 1's on the diagonal, so is L^{-1} .

A triangular matrix is invertible if and only if no diagonal entries are zero.

Here L has 1's so L^{-1} also has 1's. Use the Gauss-Jordan method to construct L^{-1} . Start by subtracting multiples of pivot rows from rows *below*. Normally this gets us halfway to the inverse, but for L it gets us all the way. L^{-1} appears on the right when I appears on the left. Notice how L^{-1} contains 11, from 3 times 5 minus 4.

Gauss-Jordan on triangular L

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 3 & 1 & 0 & 0 & 1 & 0 \\ 4 & 5 & 1 & 0 & 0 & 1 \end{bmatrix} = [L \ I]$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & 1 & 0 \\ 0 & 5 & 1 & -4 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \text{(3 times row 1 from row 2)} \\ \text{(4 times row 1 from row 3)} \\ \text{(then 5 times row 2 from row 3)} \end{array}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & 1 & 0 \\ 0 & 0 & 1 & 11 & -5 & 1 \end{bmatrix} = [I \ L^{-1}].$$

L goes to I by a product of elimination matrices $E_{32}E_{31}E_{21}$. So that product is L^{-1} . All pivots are 1's (a full set). L^{-1} is lower triangular, with the strange entry "11".

That 11 does not appear to spoil 3, 4, 5 in the good order $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = L$.

■ REVIEW OF THE KEY IDEAS ■

1. The inverse matrix gives $AA^{-1} = I$ and $A^{-1}A = I$.
2. A is invertible if and only if it has n pivots (row exchanges allowed).
3. If $Ax = 0$ for a nonzero vector x , then A has no inverse.
4. The inverse of AB is the reverse product $B^{-1}A^{-1}$. And $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$.
5. The Gauss-Jordan method solves $AA^{-1} = I$ to find the n columns of A^{-1} . The augmented matrix $[A \ I]$ is row-reduced to $[I \ A^{-1}]$.

■ WORKED EXAMPLES ■

2.5 A The inverse of a triangular **difference matrix** A is a triangular **sum matrix** S :

$$[A \ I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{array} \right] = [I \ A^{-1}] = [I \ \text{sum matrix}].$$

If I change a_{13} to -1 , then all rows of A add to zero. The equation $Ax = 0$ will now have the nonzero solution $x = (1, 1, 1)$. A clear signal: ***This new A can't be inverted.***

2.5 B Three of these matrices are invertible, and three are singular. Find the inverse when it exists. Give reasons for noninvertibility (zero determinant, too few pivots, nonzero solution to $Ax = 0$) for the other three. The matrices are in the order A, B, C, D, S, E :

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 \\ 8 & 7 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 0 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Solution

$$B^{-1} = \frac{1}{4} \begin{bmatrix} 7 & -3 \\ -8 & 4 \end{bmatrix} \quad C^{-1} = \frac{1}{36} \begin{bmatrix} 0 & 6 \\ 6 & -6 \end{bmatrix} \quad S^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

A is not invertible because its determinant is $4 \cdot 6 - 3 \cdot 8 = 24 - 24 = 0$. D is not invertible because there is only one pivot; the second row becomes zero when the first row is subtracted. E is not invertible because a combination of the columns (the second column minus the first column) is zero—in other words $Ex = 0$ has the solution $x = (-1, 1, 0)$.

Of course all three reasons for noninvertibility would apply to each of A, D, E .

2.5 C Apply the Gauss-Jordan method to invert this triangular “Pascal matrix” L . You see **Pascal’s triangle**—adding each entry to the entry on its left gives the entry below. The entries of L are “binomial coefficients”. The next row would be 1, 4, 6, 4, 1.

$$\text{Triangular Pascal matrix } L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} = \text{abs}(\text{pascal}(4, 1))$$

Solution Gauss-Jordan starts with $[L \ I]$ and produces zeros by subtracting row 1:

$$[L \ I] = \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 3 & 3 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 3 & 3 & 1 & -1 & 0 & 0 & 1 \end{array} \right].$$

The next stage creates zeros below the second pivot, using multipliers 2 and 3. Then the last stage subtracts 3 times the new row 3 from the new row 4:

$$\rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 1 & 2 & -3 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 1 \end{array} \right] = [I \ L^{-1}].$$

All the pivots were 1! So we didn’t need to divide rows by pivots to get I . The inverse matrix L^{-1} looks like L itself, except odd-numbered diagonals have minus signs.

The same pattern continues to n by n Pascal matrices, L^{-1} has “alternating diagonals”.

Problem Set 2.5

- 1 Find the inverses (directly or from the 2 by 2 formula) of A, B, C :

$$A = \begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & 2 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}.$$

- 2 For these “permutation matrices” find P^{-1} by trial and error (with 1’s and 0’s):

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

- 3 Solve for the first column (x, y) and second column (t, z) of A^{-1} :

$$\begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 10 & 20 \\ 20 & 50 \end{bmatrix} \begin{bmatrix} t \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

- 4 Show that $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ is not invertible by trying to solve $AA^{-1} = I$ for column 1 of A^{-1} :

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \left(\text{For a different } A, \text{ could column 1 of } A^{-1} \right. \\ \left. \text{be possible to find but not column 2?} \right)$$

- 5 Find an upper triangular U (not diagonal) with $U^2 = I$ which gives $U = U^{-1}$.

- 6 (a) If A is invertible and $AB = AC$, prove quickly that $B = C$.

(b) If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, find two different matrices such that $AB = AC$.

- 7 (Important) If A has row 1 + row 2 = row 3, show that A is not invertible:

(a) Explain why $Ax = (1, 0, 0)$ cannot have a solution.

(b) Which right sides (b_1, b_2, b_3) might allow a solution to $Ax = b$?

(c) What happens to row 3 in elimination?

- 8 If A has column 1 + column 2 = column 3, show that A is not invertible:

(a) Find a nonzero solution x to $Ax = 0$. The matrix is 3 by 3.

(b) Elimination keeps column 1 + column 2 = column 3. Explain why there is no third pivot.

- 9 Suppose A is invertible and you exchange its first two rows to reach B . Is the new matrix B invertible and how would you find B^{-1} from A^{-1} ?

- 10 Find the inverses (in any legal way) of

$$A = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 4 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 2 & 0 & 0 \\ 4 & 3 & 0 & 0 \\ 0 & 0 & 6 & 5 \\ 0 & 0 & 7 & 6 \end{bmatrix}.$$

- 11 (a) Find invertible matrices A and B such that $A + B$ is not invertible.
 (b) Find singular matrices A and B such that $A + B$ is invertible.
- 12 If the product $C = AB$ is invertible (A and B are square), then A itself is invertible. Find a formula for A^{-1} that involves C^{-1} and B .
- 13 If the product $M = ABC$ of three square matrices is invertible, then B is invertible. (So are A and C .) Find a formula for B^{-1} that involves M^{-1} and A and C .
- 14 If you add row 1 of A to row 2 to get B , how do you find B^{-1} from A^{-1} ?

Notice the order. The inverse of $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} A \end{bmatrix}$ is _____.

- 15 Prove that a matrix with a column of zeros cannot have an inverse.
- 16 Multiply $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ times $\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. What is the inverse of each matrix if $ad \neq bc$?
- 17 (a) What 3 by 3 matrix E has the same effect as these three steps? Subtract row 1 from row 2, subtract row 1 from row 3, then subtract row 2 from row 3.
 (b) What single matrix L has the same effect as these three reverse steps? Add row 2 to row 3, add row 1 to row 3, then add row 1 to row 2.
- 18 If B is the inverse of A^2 , show that AB is the inverse of A .
- 19 Find the numbers a and b that give the inverse of $5 * \text{eye}(4) - \text{ones}(4,4)$:

$$\begin{bmatrix} 4 & -1 & -1 & -1 \\ -1 & 4 & -1 & -1 \\ -1 & -1 & 4 & -1 \\ -1 & -1 & -1 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} a & b & b & b \\ b & a & b & b \\ b & b & a & b \\ b & b & b & a \end{bmatrix}.$$

What are a and b in the inverse of $6 * \text{eye}(5) - \text{ones}(5,5)$?

- 20 Show that $A = 4 * \text{eye}(4) - \text{ones}(4,4)$ is *not* invertible: Multiply $A * \text{ones}(4,1)$.
- 21 There are sixteen 2 by 2 matrices whose entries are 1's and 0's. How many of them are invertible?

Questions 22–28 are about the Gauss-Jordan method for calculating A^{-1} .

- 22 Change I into A^{-1} as you reduce A to I (by row operations):

$$[A \ I] = \begin{bmatrix} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{bmatrix} \quad \text{and} \quad [A \ I] = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{bmatrix}$$

- 23 Follow the 3 by 3 text example but with plus signs in A . Eliminate above and below the pivots to reduce $[A \ I]$ to $[I \ A^{-1}]$:

$$[A \ I] = \begin{bmatrix} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}.$$

- 24** Use Gauss-Jordan elimination on $[U \ I]$ to find the upper triangular U^{-1} :

$$UU^{-1} = I \quad \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- 25** Find A^{-1} and B^{-1} (if they exist) by elimination on $[A \ I]$ and $[B \ I]$:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

- 26** What three matrices E_{21} and E_{12} and D^{-1} reduce $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ to the identity matrix? Multiply $D^{-1}E_{12}E_{21}$ to find A^{-1} .

- 27** Invert these matrices A by the Gauss-Jordan method starting with $[A \ I]$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}.$$

- 28** Exchange rows and continue with Gauss-Jordan to find A^{-1} :

$$[A \ I] = \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 2 & 0 & 1 \end{bmatrix}.$$

- 29** True or false (with a counterexample if false and a reason if true):

- (a) A 4 by 4 matrix with a row of zeros is not invertible.
- (b) Every matrix with 1's down the main diagonal is invertible.
- (c) If A is invertible then A^{-1} and A^2 are invertible.

- 30** For which three numbers c is this matrix not invertible, and why not?

$$A = \begin{bmatrix} 2 & c & c \\ c & c & c \\ 8 & 7 & c \end{bmatrix}.$$

- 31** Prove that A is invertible if $a \neq 0$ and $a \neq b$ (find the pivots or A^{-1}):

$$A = \begin{bmatrix} a & b & b \\ a & a & b \\ a & a & a \end{bmatrix}.$$

- 32 This matrix has a remarkable inverse. Find A^{-1} by elimination on $[A \ I]$. Extend to a 5 by 5 “alternating matrix” and guess its inverse; then multiply to confirm.

$$\text{Invert } A = \begin{bmatrix} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \text{ and solve } Ax = (1, 1, 1, 1).$$

- 33 Suppose the matrices P and Q have the same rows as I but in any order. They are “permutation matrices”. Show that $P - Q$ is singular by solving $(P - Q)x = 0$.
- 34 Find and check the inverses (assuming they exist) of these block matrices:

$$\begin{bmatrix} I & 0 \\ C & I \end{bmatrix} \quad \begin{bmatrix} A & 0 \\ C & D \end{bmatrix} \quad \begin{bmatrix} 0 & I \\ I & D \end{bmatrix}.$$

- 35 Could a 4 by 4 matrix A be invertible if every row contains the numbers 0, 1, 2, 3 in some order? What if every row of B contains 0, 1, 2, -3 in some order?
- 36 In the Worked Example 2.5 C, the triangular Pascal matrix L has an inverse with “alternating diagonals”. Check that this L^{-1} is $DL D$, where the diagonal matrix D has alternating entries 1, -1 , 1, -1 . Then $LDL D = I$, so what is the inverse of $LD = \text{pascal}(4, 1)$?
- 37 The Hilbert matrices have $H_{ij} = 1/(i + j - 1)$. Ask MATLAB for the exact 6 by 6 inverse `invhilb(6)`. Then ask it to compute `inv(hilb(6))`. How can these be different, when the computer never makes mistakes?
- 38 (a) Use `inv(P)` to invert MATLAB’s 4 by 4 symmetric matrix $P = \text{pascal}(4)$.
(b) Create Pascal’s lower triangular $L = \text{abs}(\text{pascal}(4, 1))$ and test $P = LL^T$.
- 39 If $A = \text{ones}(4)$ and $b = \text{rand}(4, 1)$, how does MATLAB tell you that $Ax = b$ has no solution? For the special $b = \text{ones}(4, 1)$, which solution to $Ax = b$ is found by $A \backslash b$?

Challenge Problems

- 40 (Recommended) A is a 4 by 4 matrix with 1’s on the diagonal and $-a, -b, -c$ on the diagonal above. Find A^{-1} for this bidiagonal matrix.
- 41 Suppose E_1, E_2, E_3 are 4 by 4 identity matrices, except E_1 has a, b, c in column 1 and E_2 has d, e in column 2 and E_3 has f in column 3 (below the 1’s). Multiply $L = E_1 E_2 E_3$ to show that all these nonzeros are copied into L .
 $E_1 E_2 E_3$ is in the *opposite* order from elimination (because E_3 is acting first). But $E_1 E_2 E_3 = L$ is in the *correct* order to invert elimination and recover A .

- 42** Direct multiplications 1–4 give $MM^{-1} = I$, and I would recommend doing #3. M^{-1} shows the change in A^{-1} (useful to know) when a matrix is subtracted from A :

$$\begin{array}{ll} 1 & M = I - uv^T \quad \text{and} \quad M^{-1} = I + uv^T / (1 - v^T u) \quad (\text{rank 1 change in } I) \\ 2 & M = A - uv^T \quad \text{and} \quad M^{-1} = A^{-1} + A^{-1} uv^T A^{-1} / (1 - v^T A^{-1} u) \\ 3 & M = I - UV \quad \text{and} \quad M^{-1} = I_n + U(I_m - VU)^{-1} V \\ 4 & M = A - UW^{-1}V \quad \text{and} \quad M^{-1} = A^{-1} + A^{-1}U(W - VA^{-1}U)^{-1}VA^{-1} \end{array}$$

The Woodbury-Morrison formula 4 is the “matrix inversion lemma” in engineering. The *Kalman filter* for solving block tridiagonal systems uses formula 4 at each step. The four matrices M^{-1} are in diagonal blocks when inverting these block matrices (v^T is 1 by n , u is n by 1, V is m by n , U is n by m).

$$\begin{bmatrix} I & u \\ v^T & 1 \end{bmatrix} \quad \begin{bmatrix} A & u \\ v^T & 1 \end{bmatrix} \quad \begin{bmatrix} I_n & U \\ V & I_m \end{bmatrix} \quad \begin{bmatrix} A & U \\ V & W \end{bmatrix}$$

- 43** Second difference matrices have beautiful inverses if they start with $T_{11} = 1$ (instead of $K_{11} = 2$). Here is the 3 by 3 tridiagonal matrix T and its inverse:

$$T_{11} = 1 \quad T = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \quad T^{-1} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

One approach is Gauss-Jordan elimination on $[T \ I]$. That seems too mechanical. I would rather write T as the product of first differences L times U . The inverses of L and U in Worked Example 2.5 A are **sum matrices**, so here are T and T^{-1} :

$$LU = \begin{bmatrix} 1 & & & \\ -1 & 1 & & \\ 0 & -1 & 1 & \\ \text{difference} & & & \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ & 1 & -1 \\ & & 1 \\ \text{difference} & & & \end{bmatrix} \quad U^{-1}L^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & 1 \\ \text{sum} & & & \end{bmatrix} \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 1 & 1 & \\ \text{sum} & & & \end{bmatrix}$$

Question. (4 by 4) What are the pivots of T ? What is its 4 by 4 inverse? The reverse order UL gives what matrix T^* ? What is the inverse of T^* ?

- 44** Here are two more difference matrices, both important. *But are they invertible?*

$$\text{Cyclic } C = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{bmatrix} \quad \text{Free ends } F = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

One test is elimination—the fourth pivot fails. Another test is the determinant, we don’t want that. The best way is much faster, and independent of matrix size:

Produce $x \neq 0$ so that $Cx = 0$. Do the same for $Fx = 0$. Not invertible.

Show how both equations $Cx = b$ and $Fx = b$ lead to $0 = b_1 + b_2 + \cdots + b_n$. There is no solution for other b .

- 45** *Elimination for a 2 by 2 block matrix:* When you multiply the first block row by CA^{-1} and subtract from the second row, the “Schur complement” S appears:

$$\begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \quad \begin{array}{l} A \text{ and } D \text{ are square} \\ S = D - CA^{-1}B. \end{array}$$

Multiply on the right to subtract $A^{-1}B$ times block column 1 from block column 2.

$$\begin{bmatrix} A & B \\ 0 & S \end{bmatrix} \begin{bmatrix} I & -A^{-1}B \\ 0 & I \end{bmatrix} = ? \quad \text{Find } S \text{ for } \begin{bmatrix} A & B \\ C & I \end{bmatrix} = \begin{bmatrix} 2 & 3 & 3 \\ 4 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}.$$

The block pivots are A and S . If they are invertible, so is $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$.

- 46** How does the identity $A(I + BA) = (I + AB)A$ connect the inverses of $I + BA$ and $I + AB$? Those are both invertible or both singular: not obvious.

2.6 Elimination = Factorization: $A = LU$

Students often say that mathematics courses are too theoretical. Well, not this section. It is almost purely practical. The goal is to describe Gaussian elimination in the most useful way. Many key ideas of linear algebra, when you look at them closely, are really *factorizations* of a matrix. The original matrix A becomes the product of two or three special matrices. The first factorization—also the most important in practice—comes now from elimination. ***The factors L and U are triangular matrices. The factorization that comes from elimination is $A = LU$.***

We already know U , the upper triangular matrix with the pivots on its diagonal. The elimination steps take A to U . We will show how reversing those steps (taking U back to A) is achieved by a lower triangular L . ***The entries of L are exactly the multipliers ℓ_{ij} —which multiplied the pivot row j when it was subtracted from row i .***

Start with a 2 by 2 example. The matrix A contains 2, 1, 6, 8. The number to eliminate is 6. ***Subtract 3 times row 1 from row 2.*** That step is E_{21} in the forward direction with multiplier $\ell_{21} = 3$. The return step from U to A is $L = E_{21}^{-1}$ (an addition using +3):

$$\text{Forward from } A \text{ to } U : \quad E_{21}A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 6 & 8 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix} = U$$

$$\text{Back from } U \text{ to } A : \quad E_{21}^{-1}U = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & 8 \end{bmatrix} = A.$$

The second line is our factorization $LU = A$. Instead of E_{21}^{-1} we write L . Move now to larger matrices with many E 's. ***Then L will include all their inverses.***

Each step from A to U multiplies by a matrix E_{ij} to produce zero in the (i, j) position. To keep this clear, we stay with the most frequent case—***when no row exchanges are involved.*** If A is 3 by 3, we multiply by E_{21} and E_{31} and E_{32} . The multipliers ℓ_{ij} produce zeros in the $(2, 1)$ and $(3, 1)$ and $(3, 2)$ positions—all below the diagonal. Elimination ends with the upper triangular U .

Now move those E 's onto the other side, *where their inverses multiply U :*

$$(E_{32}E_{31}E_{21})A = U \quad \text{becomes} \quad A = (E_{21}^{-1}E_{31}^{-1}E_{32}^{-1})U \quad \text{which is} \quad A = LU. \quad (1)$$

The inverses go in opposite order, as they must. That product of three inverses is L . ***We have reached $A = LU$.*** Now we stop to understand it.

Explanation and Examples

First point: Every inverse matrix E^{-1} is *lower triangular*. Its off-diagonal entry is ℓ_{ij} , to undo the subtraction produced by $-\ell_{ij}$. The main diagonals of E and E^{-1} contain 1's. Our example above had $\ell_{21} = 3$ and $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ and $L = E^{-1} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$.

Second point: Equation (1) shows a lower triangular matrix (the product of the E_{ij}) multiplying A . It also shows all the E_{ij}^{-1} multiplying U to bring back A . ***This lower triangular product of inverses is L .***

One reason for working with the inverses is that we want to factor A , not U . The “inverse form” gives $A = LU$. Another reason is that we get something extra, almost more than we deserve. This is the third point, showing that L is exactly right.

Third point: Each multiplier ℓ_{ij} goes directly into its i, j position—*unchanged*—in the product of inverses which is L . Usually matrix multiplication will mix up all the numbers. Here that doesn’t happen. The order is right for the inverse matrices, to keep the ℓ ’s unchanged. The reason is given below in equation (3).

Since each E^{-1} has 1’s down its diagonal, the final good point is that L does too.

($A = LU$) This is elimination without row exchanges. The upper triangular U has the pivots on its diagonal. The lower triangular L has all 1’s on its diagonal. *The multipliers ℓ_{ij} are below the diagonal of L .*

Example 1 Elimination subtracts $\frac{1}{2}$ times row 1 from row 2. The last step subtracts $\frac{2}{3}$ times row 2 from row 3. The lower triangular L has $\ell_{21} = \frac{1}{2}$ and $\ell_{32} = \frac{2}{3}$. Multiplying LU produces A :

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 0 \\ 0 & \frac{3}{2} & 1 \\ 0 & 0 & \frac{4}{3} \end{bmatrix} = LU.$$

The (3, 1) multiplier is zero because the (3, 1) entry in A is zero. No operation needed.

Example 2 Change the top left entry from 2 to 1. The pivots all become 1. The multipliers are all 1. That pattern continues when A is 4 by 4:

$$\text{Special pattern} \quad A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 0 & 1 & 1 & \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 0 \\ & 1 & 1 & 0 \\ & & 1 & 1 \\ & & & 1 \end{bmatrix}.$$

These LU examples are showing something extra, which is very important in practice. Assume no row exchanges. When can we predict zeros in L and U ?

When a row of A starts with zeros, so does that row of L .

When a column of A starts with zeros, so does that column of U .

If a row starts with zero, we don’t need an elimination step. L has a zero, which saves computer time. Similarly, zeros at the *start* of a column survive into U . But please realize: Zeros in the *middle* of a matrix are likely to be filled in, while elimination sweeps forward. We now explain why L has the multipliers ℓ_{ij} in position, with no mix-up.

The key reason why A equals LU : Ask yourself about the pivot rows that are subtracted from lower rows. Are they the original rows of A ? *No*, elimination probably changed them. Are they rows of U ? *Yes*, the pivot rows never change again. When computing the third

row of U , we subtract multiples of earlier rows of U (*not rows of A !*):

$$\text{Row 3 of } U = (\text{Row 3 of } A) - \ell_{31}(\text{Row 1 of } U) - \ell_{32}(\text{Row 2 of } U). \quad (2)$$

Rewrite this equation to see that the row $[\ell_{31} \ \ell_{32} \ 1]$ is multiplying U :

$$(\text{Row 3 of } A) = \ell_{31}(\text{Row 1 of } U) + \ell_{32}(\text{Row 2 of } U) + 1(\text{Row 3 of } U). \quad (3)$$

This is exactly row 3 of $A = LU$. That row of L holds $\ell_{31}, \ell_{32}, 1$. All rows look like this, whatever the size of A . With no row exchanges, we have $A = LU$.

Better balance The LU factorization is “unsymmetric” because U has the pivots on its diagonal where L has 1’s. This is easy to change. **Divide U by a diagonal matrix D that contains the pivots.** That leaves a new matrix with 1’s on the diagonal:

$$\text{Split } U \text{ into } \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_n \end{bmatrix} \begin{bmatrix} 1 & u_{12}/d_1 & u_{13}/d_1 & \cdot \\ & 1 & u_{23}/d_2 & \cdot \\ & & \ddots & \vdots \\ & & & 1 \end{bmatrix}.$$

It is convenient (but a little confusing) to keep the same letter U for this new upper triangular matrix. It has 1’s on the diagonal (like L). Instead of the normal LU , the new form has D in the middle: **Lower triangular L times diagonal D times upper triangular U .**

The triangular factorization can be written $A = LU$ or $A = LDU$.

Whenever you see LDU , it is understood that U has 1’s on the diagonal. *Each row is divided by its first nonzero entry—the pivot.* Then L and U are treated evenly in LDU :

$$\begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 8 \\ 0 & 5 \end{bmatrix} \text{ splits further into } \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & \\ & 5 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}. \quad (4)$$

The pivots 2 and 5 went into D . Dividing the rows by 2 and 5 left the rows $[1 \ 4]$ and $[0 \ 1]$ in the new U with diagonal ones. The multiplier 3 is still in L .

My own lectures sometimes stop at this point. The next paragraphs show how elimination codes are organized, and how long they take. If MATLAB (or any software) is available, you can measure the computing time by just counting the seconds.

One Square System = Two Triangular Systems

The matrix L contains our memory of Gaussian elimination. It holds the numbers that multiplied the pivot rows, before subtracting them from lower rows. When do we need this record and how do we use it in solving $Ax = b$?

We need L as soon as there is a *right side* b . The factors L and U were completely decided by the left side (the matrix A). On the right side of $Ax = b$, we use L^{-1} and then U^{-1} . That *Solve* step deals with two triangular matrices.

- 1 **Factor** (into L and U , by elimination on the left side matrix A)
- 2 **Solve** (forward elimination on b using L , then back substitution for x using U).

Earlier, we worked on A and b at the same time. No problem with that—just augment to $[A \ b]$. But most computer codes keep the two sides separate. The memory of elimination is held in L and U , to process b whenever we want to. The User's Guide to LAPACK remarks that "This situation is so common and the savings are so important that no provision has been made for solving a single system with just one subroutine."

How does **Solve** work on b ? First, apply forward elimination to the right side (the multipliers are stored in L , use them now). This changes b to a new right side c . *We are really solving $Lc = b$* . Then back substitution solves $Ux = c$ as always. The original system $Ax = b$ is factored into *two triangular systems*:

$$\text{Forward and backward Solve } Lc = b \text{ and then solve } Ux = c. \quad (5)$$

To see that x is correct, multiply $Ux = c$ by L . Then $LUx = Lc$ is just $Ax = b$.

To emphasize: There is *nothing new* about those steps. This is exactly what we have done all along. We were really solving the triangular system $Lc = b$ as elimination went forward. Then back substitution produced x . An example shows what we actually did.

Example 3 Forward elimination (downward) on $Ax = b$ ends at $Ux = c$:

$$Ax = b \quad \begin{array}{l} u + 2v = 5 \\ 4u + 9v = 21 \end{array} \quad \text{becomes} \quad \begin{array}{l} u + 2v = 5 \\ v = 1 \end{array} \quad Ux = c$$

The multiplier was 4, which is saved in L . The right side used it to change 21 to 1:

$$Lc = b \quad \text{The lower triangular system} \quad \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} c \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 21 \end{bmatrix} \quad \text{gave} \quad c = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

$$Ux = c \quad \text{The upper triangular system} \quad \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ x \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \quad \text{gives} \quad x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

L and U can go into the n^2 storage locations that originally held A (now forgettable).

The Cost of Elimination

A very practical question is cost—or computing time. We can solve 1000 equations on a PC. What if $n = 100,000$? (*Not if A is dense.*) Large systems come up all the time in scientific computing, where a three-dimensional problem can easily lead to a million unknowns. We can let the calculation run overnight, but we can't leave it for 100 years.

The first stage of elimination, on column 1, produces zeros below the first pivot. To find each new entry below the pivot row requires one multiplication and one subtraction. *We will count this first stage as n^2 multiplications and n^2 subtractions.* It is actually less, $n^2 - n$, because row 1 does not change.

The next stage clears out the second column below the second pivot. The working matrix is now of size $n - 1$. Estimate this stage by $(n - 1)^2$ multiplications and subtractions. The matrices are getting smaller as elimination goes forward. The rough count to reach U is the sum of squares $n^2 + (n - 1)^2 + \cdots + 2^2 + 1^2$.

There is an exact formula $\frac{1}{3}n(n + \frac{1}{2})(n + 1)$ for this sum of squares. When n is large, the $\frac{1}{2}$ and the 1 are not important. *The number that matters is $\frac{1}{3}n^3$.* The sum of squares is like the integral of x^2 ! The integral from 0 to n is $\frac{1}{3}n^3$:

Elimination on A requires about $\frac{1}{3}n^3$ multiplications and $\frac{1}{3}n^3$ subtractions.

What about the right side b ? Going forward, we subtract multiples of b_1 from the lower components $b_2, \dots, b_n$. This is $n - 1$ steps. The second stage takes only $n - 2$ steps, because b_1 is not involved. The last stage of forward elimination takes one step.

Now start back substitution. Computing x_n uses one step (divide by the last pivot). The next unknown uses two steps. When we reach x_1 it will require n steps ($n - 1$ substitutions of the other unknowns, then division by the first pivot). The total count on the right side, from b to c to x —forward to the bottom and back to the top—is exactly n^2 :

$$[(n - 1) + (n - 2) + \cdots + 1] + [1 + 2 + \cdots + (n - 1) + n] = n^2. \quad (6)$$

To see that sum, pair off $(n - 1)$ with 1 and $(n - 2)$ with 2. The pairings leave n terms, each equal to n . That makes n^2 . The right side costs a lot less than the left side!

Solve Each right side needs n^2 multiplications and n^2 subtractions.

A **band matrix** B has only w nonzero diagonals below and also above its main diagonal. The zero entries outside the band stay zero in elimination (zeros in L and U). Clearing out the first column needs w^2 multiplications and subtractions (w zeros to be produced below the pivot, each one using a pivot row of length w). Then clearing out all n columns, to reach U , needs no more than nw^2 . This saves a lot of time:

Band matrices	Factor	change $\frac{1}{3}n^3$ to nw^2	Solve	change n^2 to $2nw$
----------------------	---------------	-----------------------------------	--------------	-----------------------

Here are codes to factor A into LU and to solve $Ax = b$. The Teaching code **slu** stops right away if a number smaller than the tolerance “*tol*” appears in a pivot position. The Teaching Codes are on web.mit.edu/18.06/www. Professional codes will look down each column for the largest available pivot, to exchange rows and continue solving.

MATLAB’s backslash command $x = A \backslash b$ combines **Factor** and **Solve** to reach x .

```

function [L,U] = slu(A)
%   Square LU factorization with no row exchanges!
[n,n] = size(A);  tol = 1.e - 6;
for k = 1 : n
    if abs(A(k,k)) < tol
        end      % Cannot proceed without a row exchange: stop
        L(k,k) = 1;
        for i = k + 1 : n
            L(i,k) = A(i,k)/A(k,k);  % Multipliers for column k are put into L
            for j = k + 1 : n  % Elimination beyond row k and column k
                A(i,j) = A(i,j) - L(i,k) * A(k,j);  % Matrix still called A
            end
        end
        for j = k : n
            U(k,j) = A(k,j);  % row k is settled, now name it U
        end
    end
end

function x = slv(A,b)
%   Solve Ax = b using L and U from slu(A).
[L,U] = slu(A); s = 0;  % No row exchanges!
for k = 1 : n  % Forward elimination to solve Lc = b
    for j = 1 : k - 1
        s = s + L(k,j) * c(j);  % Add L times earlier c(j) before c(k)
    end
    c(k) = b(k) - s; s = 0;  % Find c(k) and reset s for next k
end
for k = n : -1 : 1  % Going backwards from x(n) to x(1)
    for j = k + 1 : n  % Back substitution
        t = t + U(k,j) * x(j);  % U times later x(j)
    end
    x(k) = (c(k) - t)/U(k,k);  % Divide by pivot
end
x = x';  % Transpose to column vector

```

How long does it take to solve $Ax = b$? For a random matrix of order $n = 1000$, a typical time is 1 second. See web.mit.edu/18.06 and math.mit.edu/linearalgebra for the times in MATLAB, Maple, Mathematica, SciLab, Python, and R. The time is multiplied by about 8 when n is multiplied by 2. For professional codes go to netlib.org.

According to this n^3 rule, matrices that are 10 times as large (order 10,000) will take a thousand seconds. Matrices of order 100,000 will take a million seconds. This is too expensive without a supercomputer, but remember that these matrices are full. Most matrices in practice are sparse (many zero entries). In that case $A = LU$ is much faster.

For tridiagonal matrices of order 10,000, storing only the nonzeros, solving $Ax = b$ is a breeze. Provided the code recognizes that A is tridiagonal.

■ REVIEW OF THE KEY IDEAS ■

1. Gaussian elimination (with no row exchanges) factors A into L times U .
2. The lower triangular L contains the numbers ℓ_{ij} that multiply pivot rows, going from A to U . The product LU adds those rows back to recover A .
3. On the right side we solve $Lc = b$ (forward) and $Ux = c$ (backward).
4. **Factor** : There are $\frac{1}{3}(n^3 - n)$ multiplications and subtractions on the left side.
5. **Solve** : There are n^2 multiplications and subtractions on the right side.
6. For a band matrix, change $\frac{1}{3}n^3$ to nw^2 and change n^2 to $2wn$.

■ WORKED EXAMPLES ■

2.6 A The lower triangular Pascal matrix L contains the famous “Pascal triangle”. Gauss-Jordan found its inverse in the worked example 2.5 C. This problem connects L to the *symmetric* Pascal matrix P and the upper triangular U . The symmetric P has Pascal’s triangle tilted, so each entry is the sum of the entry above and the entry to the left. The n by n symmetric P is `pascal(n)` in MATLAB.

Problem: Establish the amazing lower-upper factorization $P = LU$.

$$\text{pascal}(4) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = LU.$$

Then predict and check the next row and column for 5 by 5 Pascal matrices.

Solution You could multiply LU to get P . Better to start with the symmetric P and reach the upper triangular U by elimination:

$$P = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 3 & 6 & 10 \\ 1 & 4 & 10 & 20 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 2 & 5 & 9 \\ 0 & 3 & 9 & 19 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 3 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U.$$

The multipliers ℓ_{ij} that entered these steps go perfectly into L . Then $P = LU$ is a particularly neat example. Notice that every pivot is 1 on the diagonal of U .

The next section will show how symmetry produces a special relationship between the triangular L and U . For Pascal, U is the “transpose” of L .

You might expect the MATLAB command `lu(pascal(4))` to produce these L and U . That doesn't happen because the `lu` subroutine chooses the largest available pivot in each column. The second pivot will change from 1 to 3. But a "Cholesky factorization" does no row exchanges: $U = \text{chol}(\text{pascal}(4))$

The full proof of $P = LU$ for all Pascal sizes is quite fascinating. The paper "*Pascal Matrices*" is on the course web page web.mit.edu/18.06 which is also available through MIT's *OpenCourseWare* at ocw.mit.edu. These Pascal matrices have so many remarkable properties—we will see them again.

2.6 B The problem is: Solve $Px = b = (1, 0, 0, 0)$. This right side = column of I means that x will be the first column of P^{-1} . That is Gauss-Jordan, matching the columns of $PP^{-1} = I$. We already know the Pascal matrices L and U as factors of P :

$$\text{Two triangular systems} \quad Lc = b \text{ (forward)} \quad Ux = c \text{ (back).}$$

Solution The lower triangular system $Lc = b$ is solved *top to bottom*:

$$\begin{array}{rclcl} c_1 & = & 1 & & c_1 = +1 \\ c_1 + c_2 & = & 0 & & c_2 = -1 \\ c_1 + 2c_2 + c_3 & = & 0 & \text{gives} & c_3 = +1 \\ c_1 + 3c_2 + 3c_3 + c_4 & = & 0 & & c_4 = -1 \end{array}$$

Forward elimination is multiplication by L^{-1} . It produces the upper triangular system $Ux = c$. The solution x comes as always by back substitution, *bottom to top*:

$$\begin{array}{rclcl} x_1 + x_2 + x_3 + x_4 & = & 1 & & x_1 = +4 \\ x_2 + 2x_3 + 3x_4 & = & -1 & & x_2 = -6 \\ x_3 + 3x_4 & = & 1 & \text{gives} & x_3 = +4 \\ x_4 & = & -1 & & x_4 = -1 \end{array}$$

I see a pattern in that x , but I don't know where it comes from. Try `inv(pascal(4))`.

Problem Set 2.6

Problems 1–14 compute the factorization $A = LU$ (and also $A = LDU$).

- 1 (Important) Forward elimination changes $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}x = b$ to a triangular $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}x = c$:

$$\begin{array}{rcl} \begin{array}{l} x + y = 5 \\ x + 2y = 7 \end{array} & \longrightarrow & \begin{array}{l} x + y = 5 \\ y = 2 \end{array} \end{array} \quad \begin{array}{l} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 1 & 2 & 7 \end{array} \right] \longrightarrow \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & 2 \end{array} \right] \end{array}$$

That step subtracted $\ell_{21} = \underline{\hspace{1cm}}$ times row 1 from row 2. The reverse step *adds* ℓ_{21} times row 1 to row 2. The matrix for that reverse step is $L = \underline{\hspace{1cm}}$. Multiply this L times the triangular system $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}x_1 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ to get $\underline{\hspace{1cm}} = \underline{\hspace{1cm}}$. In letters, L multiplies $Ux = c$ to give $\underline{\hspace{1cm}}$.

- 2 Write down the 2 by 2 triangular systems $Lc = b$ and $Ux = c$ from Problem 1. Check that $c = (5, 2)$ solves the first one. Find x that solves the second one.

- 3 (Move to 3 by 3) Forward elimination changes $Ax = b$ to a triangular $Ux = c$:

$$\begin{array}{rcl} x + y + z = 5 & x + y + z = 5 & x + y + z = 5 \\ x + 2y + 3z = 7 & y + 2z = 2 & y + 2z = 2 \\ x + 3y + 6z = 11 & 2y + 5z = 6 & z = 2 \end{array}$$

The equation $z = 2$ in $Ux = c$ comes from the original $x + 3y + 6z = 11$ in $Ax = b$ by subtracting $\ell_{31} = \underline{\hspace{1cm}}$ times equation 1 and $\ell_{32} = \underline{\hspace{1cm}}$ times the final equation 2. Reverse that to recover $[1 \ 3 \ 6 \ 11]$ in the last row of A and b from the final $[1 \ 1 \ 1 \ 5]$ and $[0 \ 1 \ 2 \ 2]$ and $[0 \ 0 \ 1 \ 2]$ in U and c :

$$\text{Row 3 of } [A \ b] = (\ell_{31} \text{ Row 1} + \ell_{32} \text{ Row 2} + 1 \text{ Row 3}) \text{ of } [U \ c].$$

In matrix notation this is multiplication by L . So $A = LU$ and $b = Lc$.

- 4 What are the 3 by 3 triangular systems $Lc = b$ and $Ux = c$ from Problem 3? Check that $c = (5, 2, 2)$ solves the first one. Which x solves the second one?
- 5 What matrix E puts A into triangular form $EA = U$? Multiply by $E^{-1} = L$ to factor A into LU :

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 2 \\ 6 & 3 & 5 \end{bmatrix}.$$

- 6 What two elimination matrices E_{21} and E_{32} put A into upper triangular form $E_{32}E_{21}A = U$? Multiply by E_{32}^{-1} and E_{21}^{-1} to factor A into $LU = E_{21}^{-1}E_{32}^{-1}U$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 5 \\ 0 & 4 & 0 \end{bmatrix}.$$

- 7 What three elimination matrices E_{21} , E_{31} , E_{32} put A into its upper triangular form $E_{32}E_{31}E_{21}A = U$? Multiply by E_{32}^{-1} , E_{31}^{-1} and E_{21}^{-1} to factor A into L times U :

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix} \quad L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}.$$

- 8 Suppose A is already lower triangular with 1's on the diagonal. Then $U = I$!

$$A = L = \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix}.$$

The elimination matrices E_{21} , E_{31} , E_{32} contain $-a$ then $-b$ then $-c$.

- (a) Multiply $E_{32}E_{31}E_{21}$ to find the single matrix E that produces $EA = I$.
- (b) Multiply $E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}$ to bring back L (nicer than E).

- 9 When zero appears in a pivot position, $A = LU$ is not possible! (We are requiring nonzero pivots in U .) Show directly why these are both impossible:

$$\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \ell & 1 \end{bmatrix} \begin{bmatrix} d & e \\ 0 & f \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ \ell & 1 & \\ m & n & 1 \end{bmatrix} \begin{bmatrix} d & e & g \\ f & h & \\ i & & \end{bmatrix}.$$

This difficulty is fixed by a row exchange. That needs a “permutation” P .

- 10 Which number c leads to zero in the second pivot position? A row exchange is needed and $A = LU$ will not be possible. Which c produces zero in the third pivot position? Then a row exchange can’t help and elimination fails:

$$A = \begin{bmatrix} 1 & c & 0 \\ 2 & 4 & 1 \\ 3 & 5 & 1 \end{bmatrix}.$$

- 11 What are L and D (the diagonal *pivot matrix*) for this matrix A ? What is U in $A = LU$ and what is the new U in $A = LDU$?

Already triangular $A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}.$

- 12 A and B are symmetric across the diagonal (because $4 = 4$). Find their triple factorizations LDU and say how U is related to L for these symmetric matrices:

Symmetric $A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 & 0 \\ 4 & 12 & 4 \\ 0 & 4 & 0 \end{bmatrix}.$

- 13 (Recommended) Compute L and U for the symmetric matrix A :

$$A = \begin{bmatrix} a & a & a & a \\ a & b & b & b \\ a & b & c & c \\ a & b & c & d \end{bmatrix}.$$

Find four conditions on a, b, c, d to get $A = LU$ with four pivots.

- 14 This nonsymmetric matrix will have the same L as in Problem 13:

Find L and U for $A = \begin{bmatrix} a & r & r & r \\ a & b & s & s \\ a & b & c & t \\ a & b & c & d \end{bmatrix}.$

Find the four conditions on a, b, c, d, r, s, t to get $A = LU$ with four pivots.

Problems 15-16 use L and U (without needing A) to solve $Ax = b$.

- 15** Solve the triangular system $Lc = b$ to find c . Then solve $Ux = c$ to find x :

$$L = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 2 \\ 11 \end{bmatrix}.$$

For safety multiply LU and solve $Ax = b$ as usual. Circle c when you see it.

- 16** Solve $Lc = b$ to find c . Then solve $Ux = c$ to find x . **What was A ?**

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

- 17** (a) When you apply the usual elimination steps to L , what matrix do you reach?

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}.$$

(b) When you apply the same steps to I , what matrix do you get?

(c) When you apply the same steps to LU , what matrix do you get?

- 18** If $A = LDU$ and also $A = L_1 D_1 U_1$ with all factors invertible, then $L = L_1$ and $D = D_1$ and $U = U_1$. “The three factors are unique.”

Derive the equation $L_1^{-1}LD = D_1U_1U^{-1}$. Are the two sides triangular or diagonal? Deduce $L = L_1$ and $U = U_1$ (they all have diagonal 1's). Then $D = D_1$.

- 19** *Tridiagonal matrices* have zero entries except on the main diagonal and the two adjacent diagonals. Factor these into $A = LU$ and $A = LDL^T$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a & a & 0 \\ a & a+b & b \\ 0 & b & b+c \end{bmatrix}.$$

- 20** When T is tridiagonal, its L and U factors have only two nonzero diagonals. How would you take advantage of knowing the zeros in T , in a code for Gaussian elimination? Find L and U .

Tridiagonal

$$T = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 \end{bmatrix}.$$

- 21** If A and B have nonzeros in the positions marked by x , which zeros (marked by 0) stay zero in their factors L and U ?

$$A = \begin{bmatrix} x & x & x & x \\ x & x & x & 0 \\ 0 & x & x & x \\ 0 & 0 & x & x \end{bmatrix} \quad B = \begin{bmatrix} x & x & x & 0 \\ x & x & 0 & x \\ x & 0 & x & x \\ 0 & x & x & x \end{bmatrix}.$$

- 22** Suppose you eliminate upwards (almost unheard of). Use the last row to produce zeros in the last column (the pivot is 1). Then use the second row to produce zero above the second pivot. Find the factors in the unusual order $A = UL$.

Upper times lower $A = \begin{bmatrix} 5 & 3 & 1 \\ 3 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$

- 23** *Easy but important.* If A has pivots 5, 9, 3 with no row exchanges, what are the pivots for the upper left 2 by 2 submatrix A_2 (without row 3 and column 3)?

Challenge Problems

- 24** Which invertible matrices allow $A = LU$ (elimination without row exchanges)? *Good question!* Look at each of the square upper left submatrices of A .

All upper left k by k submatrices A_k must be invertible (sizes $k = 1, \dots, n$).

Explain that answer: A_k factors into _____ because $LU = \begin{bmatrix} L_k & 0 \\ * & * \end{bmatrix} \begin{bmatrix} U_k & * \\ 0 & * \end{bmatrix}.$

- 25** For the 6 by 6 second difference constant-diagonal matrix K , put the pivots and multipliers into $K = LU$. (L and U will have only two nonzero diagonals, because K has three.) Find a formula for the i, j entry of L^{-1} , by software like MATLAB using $\text{inv}(L)$ or by looking for a nice pattern.

$-1, 2, -1$ matrix $K = \begin{bmatrix} 2 & -1 & & & & \\ -1 & \cdot & \cdot & & & \\ & \cdot & \cdot & \cdot & & \\ & & \cdot & \cdot & \cdot & \\ & & & \cdot & \cdot & -1 \\ & & & & -1 & 2 \end{bmatrix} = \text{toeplitz}([2 \ -1 \ 0 \ 0 \ 0 \ 0])$

- 26** If you print K^{-1} , it doesn't look so good. But if you print $7K^{-1}$ (when K is 6 by 6), that matrix looks wonderful. Write down $7K^{-1}$ by hand, following this pattern:

- 1 Row 1 and column 1 are (6, 5, 4, 3, 2, 1).
- 2 On and above the main diagonal, row i is i times row 1.
- 3 On and below the main diagonal, column j is j times column 1.

Multiply K times that $7K^{-1}$ to produce $7I$. Here is that pattern for $n = 3$:

3 by 3 case
The determinant of this K is 4 $(K)(4K^{-1}) = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 & & \\ & 4 & \\ & & 4 \end{bmatrix}.$

2.7 Transposes and Permutations

We need one more matrix, and fortunately it is much simpler than the inverse. It is the “*transpose*” of A , which is denoted by A^T . *The columns of A^T are the rows of A .*

When A is an m by n matrix, the transpose is n by m :

$$\text{Transpose} \quad \text{If } A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \quad \text{then } A^T = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 4 \end{bmatrix}.$$

You can write the rows of A into the columns of A^T . Or you can write the columns of A into the rows of A^T . The matrix “flips over” its main diagonal. The entry in row i , column j of A^T comes from row j , column i of the original A :

$$\text{Exchange rows and columns} \quad (A^T)_{ij} = A_{ji}.$$

The transpose of a lower triangular matrix is upper triangular. (But the inverse is still lower triangular.) The transpose of A^T is A .

Note MATLAB’s symbol for the transpose of A is A' . Typing $[1 \ 2 \ 3]$ gives a row vector and the column vector is $v = [1 \ 2 \ 3]'$. To enter a matrix M with second column $w = [4 \ 5 \ 6]'$ you could define $M = [v \ w]$. Quicker to enter by rows and then transpose the whole matrix: $M = [1 \ 2 \ 3; 4 \ 5 \ 6]'$.

The rules for transposes are very direct. We can transpose $A + B$ to get $(A + B)^T$. Or we can transpose A and B separately, and then add $A^T + B^T$ —with the same result. The serious questions are about the transpose of a product AB and an inverse A^{-1} :

$$\text{Sum} \quad \text{The transpose of } A + B \text{ is } A^T + B^T. \quad (1)$$

$$\text{Product} \quad \text{The transpose of } AB \text{ is } (AB)^T = B^T A^T. \quad (2)$$

$$\text{Inverse} \quad \text{The transpose of } A^{-1} \text{ is } (A^{-1})^T = (A^T)^{-1}. \quad (3)$$

Notice especially how $B^T A^T$ comes in reverse order. For inverses, this reverse order was quick to check: $B^{-1} A^{-1}$ times AB produces I . To understand $(AB)^T = B^T A^T$, start with $(Ax)^T = x^T A^T$:

$$Ax \text{ combines the columns of } A \text{ while } x^T A^T \text{ combines the rows of } A^T.$$

It is the same combination of the same vectors! In A they are columns, in A^T they are rows. So the transpose of the column Ax is the row $x^T A^T$. That fits our formula $(Ax)^T = x^T A^T$. Now we can prove the formula $(AB)^T = B^T A^T$, when B has several columns.

If $B = [x_1 \ x_2]$ has two columns, apply the same idea to each column. The columns of AB are Ax_1 and Ax_2 . Their transposes are the rows of $B^T A^T$:

$$\text{Transposing } AB = \begin{bmatrix} Ax_1 & Ax_2 & \cdots \end{bmatrix} \text{ gives } \begin{bmatrix} x_1^T A^T \\ x_2^T A^T \\ \vdots \end{bmatrix} \text{ which is } B^T A^T. \quad (4)$$

The right answer $B^T A^T$ comes out a row at a time. Here are numbers in $(AB)^T = B^T A^T$:

$$AB = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 9 & 1 \end{bmatrix} \quad \text{and} \quad B^T A^T = \begin{bmatrix} 5 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 9 \\ 0 & 1 \end{bmatrix}.$$

The reverse order rule extends to three or more factors: $(ABC)^T$ equals $C^T B^T A^T$.

If $A = LDU$ then $A^T = U^T D^T L^T$. The pivot matrix has $D = D^T$.

Now apply this product rule to both sides of $A^{-1}A = I$. On one side, I^T is I . We confirm the rule that $(A^{-1})^T$ is the inverse of A^T , because their product is I :

Transpose of inverse $A^{-1}A = I$ is transposed to $A^T(A^{-1})^T = I$. (5)

Similarly $AA^{-1} = I$ leads to $(A^{-1})^T A^T = I$. We can invert the transpose or we can transpose the inverse. Notice especially: **A^T is invertible exactly when A is invertible.**

Example 1 The inverse of $A = \begin{bmatrix} 1 & 0 \\ 6 & 1 \end{bmatrix}$ is $A^{-1} = \begin{bmatrix} 1 & 0 \\ -6 & 1 \end{bmatrix}$. The transpose is $A^T = \begin{bmatrix} 1 & 6 \\ 0 & 1 \end{bmatrix}$.

$(A^{-1})^T$ and $(A^T)^{-1}$ are both equal to $\begin{bmatrix} 1 & -6 \\ 0 & 1 \end{bmatrix}$.

The Meaning of Inner Products

We know the dot product (inner product) of x and y . It is the sum of numbers $x_i y_i$. Now we have a better way to write $x \cdot y$, without using that unprofessional dot. Use matrix notation instead:

T is inside The dot product or inner product is $x^T y$ $(1 \times n)(n \times 1)$

T is outside The rank one product or outer product is xy^T $(n \times 1)(1 \times n)$

$x^T y$ is a number, xy^T is a matrix. Quantum mechanics would write those as $\langle x | y \rangle$ (inner) and $|x\rangle \langle y|$ (outer). I think the world is governed by linear algebra, but physics disguises it well. Here are examples where the inner product has meaning:

From mechanics Work = (Movements) (Forces) = $x^T f$

From circuits Heat loss = (Voltage drops) (Currents) = $e^T y$

From economics Income = (Quantities) (Prices) = $q^T p$

We are really close to the heart of applied mathematics, and there is one more point to explain. It is the deeper connection between inner products and the transpose of A .

We defined A^T by flipping the matrix across its main diagonal. That's not mathematics. There is a better way to approach the transpose. **A^T is the matrix that makes these two inner products equal for every x and y :**

$(Ax)^T y = x^T (A^T y)$ Inner product of Ax with y = Inner product of x with $A^T y$

Example 2 Start with $A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$ $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

On one side we have Ax multiplying y : $(x_2 - x_1)y_1 + (x_3 - x_2)y_2$

That is the same as $x_1(-y_1) + x_2(y_1 - y_2) + x_3(y_2)$. Now x is multiplying $A^T y$.

$$A^T y \text{ must be } \begin{bmatrix} -y_1 \\ y_1 - y_2 \\ y_2 \end{bmatrix} \text{ which produces } A^T = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \text{ as expected.}$$

Example 3 Will you allow me a little calculus? It is extremely important or I wouldn't leave linear algebra. (This is really linear algebra for functions $x(t)$.) **The difference matrix changes to a derivative** $A = d/dt$. Its transpose will now come from $(dx/dt, y) = (x, -dy/dt)$.

The inner product changes from a finite sum of $x_k y_k$ to an integral of $x(t)y(t)$.

Inner product of functions $x^T y = (x, y) = \int_{-\infty}^{\infty} x(t) y(t) dt$ by definition

Transpose rule $(Ax)^T y = x^T (A^T y)$ $\int_{-\infty}^{\infty} \frac{dx}{dt} y(t) dt = \int_{-\infty}^{\infty} x(t) \left(-\frac{dy}{dt}\right) dt$ shows A^T (6)

I hope you recognize “*integration by parts*”. The derivative moves from the first function $x(t)$ to the second function $y(t)$. During that move, a minus sign appears. This tells us that *the “transpose” of the derivative is minus the derivative*.

The derivative is *anti-symmetric*: $A = d/dt$ and $A^T = -d/dt$. Symmetric matrices have $A^T = A$, anti-symmetric matrices have $A^T = -A$. In some way, the 2 by 3 difference matrix above followed this pattern. The 3 by 2 matrix A^T was *minus* a difference matrix. It produced $y_1 - y_2$ in the middle component of $A^T y$ instead of the difference $y_2 - y_1$.

Symmetric Matrices

For a *symmetric matrix*, transposing A to A^T produces no change. Then $A^T = A$. Its (j, i) entry across the main diagonal equals its (i, j) entry. In my opinion, these are the most important matrices of all.

DEFINITION A *symmetric matrix* has $A^T = A$. This means that $a_{ji} = a_{ij}$.

Symmetric matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} = A^T$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 10 \end{bmatrix} = D^T$.

The inverse of a symmetric matrix is also symmetric. The transpose of A^{-1} is $(A^{-1})^T = (A^T)^{-1} = A^{-1}$. That says A^{-1} is symmetric (when A is invertible):

Symmetric inverses $A^{-1} = \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix}$ and $D^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 0.1 \end{bmatrix}$.

Now we produce symmetric matrices by *multiplying any matrix R by R^T* .